

Chapter 4: Question 1

Question 1

Using a little bit of algebra, prove that (4.2) is equivalent to (4.3). In other words, the logistic function representation and logit representation for the logistic regression model are equivalent.

Given

The logistic function in equation (4.2):

$$P(X) = \frac{e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}} \quad (4.2)$$

We want to show that this is equivalent to the odds form in equation (4.3):

$$\frac{P(X)}{1 - P(X)} = e^{\beta_0 + \beta_1 X} \quad (4.3)$$

Proof

$$P(X) = \frac{e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}} \quad (4.2)$$

Step 1: Multiply both sides of (4.2) by the denominator:

$$P(X) (1 + e^{\beta_0 + \beta_1 X}) = e^{\beta_0 + \beta_1 X}$$

Step 2: Expand the left-hand side:

$$P(X) + P(X)e^{\beta_0 + \beta_1 X} = e^{\beta_0 + \beta_1 X}$$

Step 3: Move the term $P(X)e^{\beta_0 + \beta_1 X}$ to the right-hand side:

$$P(X) = e^{\beta_0 + \beta_1 X} - P(X)e^{\beta_0 + \beta_1 X}$$

Step 4: Factor out $e^{\beta_0 + \beta_1 X}$ on the right-hand side:

$$P(X) = (1 - P(X)) e^{\beta_0 + \beta_1 X}$$

Step 5: Divide both sides by $1 - P(X)$:

$$\frac{P(X)}{1 - P(X)} = e^{\beta_0 + \beta_1 X} \quad (4.3)$$

CONCLUSION: up to here it is clearly proven that equation (4.2) is equivalent to equation (4.3).

Step 6 (Logit transformation): Apply the natural logarithm to both sides:

$$\log\left(\frac{P(X)}{1 - P(X)}\right) = \beta_0 + \beta_1 X$$

That is,

$$\text{logit}(P(X)) = \log\left(\frac{P(X)}{1 - P(X)}\right) = \beta_0 + \beta_1 X,$$

which is the **logit form of the logistic model**.